

Lab1 Report

Table No. _____ Name & Stu No. _____ Score _____ of 100

EQUIPMENTS & COMPONENTS

Digital Multimeter

Breadboards

DC Power Supply

W103 potentiometer

Wires and cables

Potentiometer adjustment tools

Resistors, marked with color code: $1k\Omega \times 3$, $1.5k\Omega$ LDR

Experiment 1, Measuring the Resistance of test cables

Touch the two probes together to create a SHORT circuit in the resistance mode. Record the resistance of the connected leads. A good cable will have an end-to-end resistance less than $1.0\ \Omega$, usually closer to $0.5\ \Omega$ or less. If the reading is within this acceptable range, jiggle the cable and confirm that the reading remains acceptable. This assures that there are no loose connections at the ends. If it reads more than an ohm, check for poor connections and ask your teacher.

1. The meter reads _____. (1pt)
2. If it does not read zero, why not? (2pts)

3. We can infer the resistance of the two test cables is _____. (1pt)

Experiment 2, Internal connections of the Breadboard

CONTINUITY TEST is used to check if two points are electrically connected.

1. If continuity exists, the beeper sounds continuously.
2. Insert the black test lead into COM terminal, and the red test lead in the terminal labeled

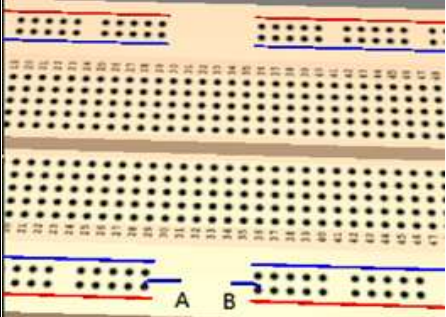
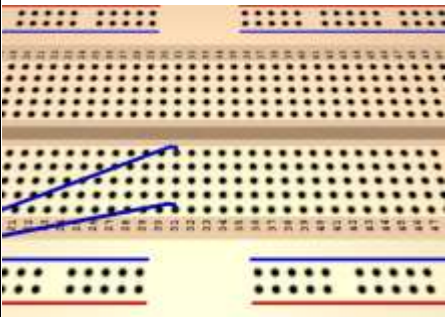
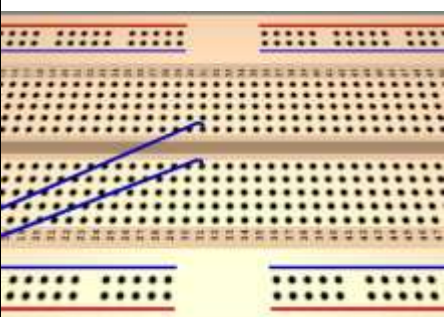


. Put the DMM in continuity mode  by pressing the button .

What did you hear? _____ Ask for help if you heard nothing. (1pt)

3. A breadboard is provided at your lab station. “Holes” that are electrically connected are **short** and said to share a common node. “Holes” that are electrically isolated from each other are **open**. Using the continuity testing function of the DMM and two jumper wires can determine which “holes” on the breadboard are connected and which are electrically isolated from each other. Connect a jumper wire to each of the alligator clips. Then put a jumper wire in one hole of a breadboard, put another jumper wire in another hole. Complete the table. (6pts, 1pt each)

Table 1

	Continuity Measurement Beeping?(Yes or NOT)	resistance between “holes”
Case A 		
Case B 		
Case C 		

Experiment 3, Resistor Color Code and Measurement of Resistance

How to use the Multimeter to measure the resistance.

1. Firstly turn the rotary switch to . Make sure power is disconnected from the circuit to be measured.
2. Insert the black test lead into COM  terminal, and the red test lead in the terminal labeled  Ω , which is the symbol for ohms.
3. Touch the tips of the probes onto each end of the resistor.
4. Read the measured resistance on the display.



Procedure

1. Select **three** 1k Ω resistors, and measure the resistance of each using the digital multimeter and compare the values with the specified value. (9pts, 1pt each)

The **percentage difference** (% Error) between the nominal value (indicated by the color code) and the measured value is calculated using the equation:

$$\text{percentage difference} = \left| \frac{\text{Nominal Value} - \text{Measured Value}}{\text{Nominal Value}} \right| \times 100\%$$

Table 2

Resistor	Measured Value(Ω)	% Error	Within Tolerance?
①-1k Ω			
②-1k Ω			
③-1k Ω			

Experiment 4, Measurement of Voltage

1. Turn the DC power supply ON. Set the DC power supply output voltage to 4 Volts, then switch the DC supply Channel ON.
2. Verify this voltage using a voltmeter.
 - 1) Warning: while measuring voltage multimeter dial SHOULD NOT be in mA (current) position. It might destroy/damage the meter
 - 2) In electronics and electrical engineering, we usually define a point in a circuit as a reference point. This reference point is known as ground (or GND) and carries a voltage of 0V. Voltage measurements are relative measurements. That is, a voltage measurement must be compared to another point in the circuit. If it is not, the measurement is meaningless. All voltage measurements are referenced with respect to ground (zero) unless otherwise noted. The conventional approach is to use the black probe so that it is “closest” to the COM (which is the “ground” reference of the circuit), and to use the red probe so that it is “closest” to the Voltage source (+4 volts in this circuit).
3. Record the voltage readings on the DMM (including sign) and the reading on the DC powersupply’s meter. (2pts, 1pt each)

Table 4

DMM	DC power supply

4. Record and explain the differences when the positive and negative leads are switched at the multimeter. (3pts)

Experiment 5, Measurement of a A Simple Electric Circuit

1. Figure 1(Figure 5@ prelab) is a voltage divider that produces an output voltage (V_{out}) that is a fraction of its input voltage (V_{in}). In this circuit, the voltage divider ratio $K = \frac{V_{out}}{V_{in}}$ is equal to the resistance ratio $\frac{R_2 // R_3}{R_1 + R_2 // R_3}$. Using these ratios $V_{out} = V_{in} \times \frac{R_2 // R_3}{R_1 + R_2 // R_3}$. Note that in figure 2, the reference voltage is ground. For the circuit shown in Figure 2, $R_1 = 1.5k\Omega$, $R_2 = R_3 = 1k\Omega$.

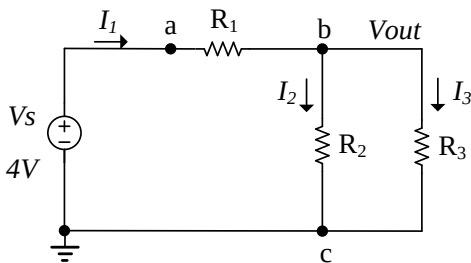


Figure 1

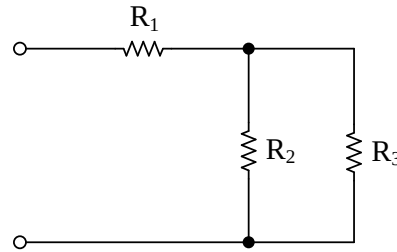


Figure 2

2. Use the DMM to measure the resistance of R_1 , R_2 and R_3 . Then connect R_1 , R_2 and R_3 as shown in Figure 1 on the breadboard, and measure the equivalent resistance R_{eq} . Record the nominal, measured and deviation values in Table 5. (18pts, 1pt each)

Table 5

	R_1	R_2	R_3	$R_2 // R_3$	R_{eq}	Voltage Divider ratio $K = \frac{R_2 // R_3}{R_1 + R_2 // R_3}$
Nominal Value	1500 Ω	1000 Ω	1000 Ω			
Tolerance (%)						
Measured Value						
% Error						

3. Build the circuit shown in Figure1 on the breadboard.
- 1) Measure the Output voltage V_{out} with the red probe on the terminal b and the black probe on the terminal c . What is the voltage reading? Now exchange the red and black probe. What is the voltage reading? Comment. (3pts, 1pts each)

--	--

- 2) Complete the Table below. (8pts, 1pt each)

Table 6

	V_{out}	V_{R1}	$V_{R1} + V_{out}$	Voltage Divider ratio $K = \frac{V_{out}}{V_{in}}$
Calculated values				
Measured values				

- 3) How does $V_{R1} + V_{out}$ compare with the applied voltage 4V? What circuit law is verified? (3pts, the first one 1pt, the last one 2pts)

--	--

- 4) Knowing the voltage drop Measured values across each resistor, use Ohm's Law to calculate the current through each resistor. (3pts, 1pts each)

--	--

4. The voltage divider ratio K of tables 5, and 6 are the same? (2pt)

--	--

5. The DMM can also be used to measure current through a device. Measuring current is different from measuring voltage or resistance. For measuring current, we must connect the meter DMM in ammeter mode in series along the branch we are measuring current in.

- 1) Set up the DMM to measure dc current.
- 2) Measure the current through R_1 . What is the current reading? Exchange the probes. What is the current reading? Comment. (7pts , the first two 2pts each, the last one 3pts)

--

- 3) Measure the current through the resistors R_2 .
- 4) Measure the current through the resistors R_3 .
- 5) Complete the Table below. (6pts, 1pt each)

Table 7

	I_1	I_2	I_3
Calculated values			
Measured values			

- 6) What is the sum of the currents I_2 and I_3 ? How does I_1 compare with $I_2 + I_3$? What circuit law is verified? (4pts, the first two 1pt, the last one 2pts,)

--

- 7) Calculate $R_{in} = V_s / I_1$ (Note: This ratio is called input resistance, which is the equivalent resistance Req “seen” by the voltage source). (2pts) Is it close to the value measured in step 1. (1pts)

--

6. We’ve already build voltage dividers with series resistors, we can also use th potentiometer’s variable resistance to construct a variable voltage divider. Replace the resistors in Figure 2 with a W103 potentiometer to achieve the same function as a voltage divider. Note that for any given position of the ball, the length of wire between the positive supply terminal and the ball serves as R_4 in the voltage divider equation, and the length of wire between the ball and the grounded terminal serves as R_5 . Therefore, by moving the ball back and forth, we change the values of R_4 and R_5 , and hence the output voltage.

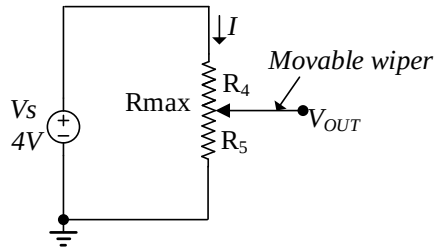


Figure 3

1. Build the circuit shown in Figure 3 on the breadboard. Apply a 4V input voltage across the end terminals of the potentiometer using the 4V fixed output on the DC power supply.
2. Read the voltage level of the wiper using the DMM while turning the knob on the potentiometer to make the voltage level of the wiper (that is V_{out} , **referenced to ground**) to be the same as the value V_{out} in table6 .

1) Record the V_{out} = _____. (1pts)

2) Calculate Voltage Divider ratio. (1pts)

K = _____

3. After getting the desired output voltage, switch off the DC power supply and remove the wires between the breadboard and the DC power supply.

1) Measure the R_4 , R_5 and R_{max} with the DMM. (3pts, 1pts each)

R_4 = _____. R_5 = _____. R_{max} = _____.

2) Calculate Voltage Divider ratio. (1pts)

K = _____

Experiment 6, Light Dependent Resistor LDR

Light Dependent Resistors (LDR), also known as photoresistors or photoconductive cells works on the principle of Photoconductivity. Photoresistors are made from semiconductor materials whose resistance changes when illuminated with light energy. One typical LDR is the CdS (Cadmium Sulfide) cell. The resistance of the CdS cell is inversely proportional to light levels. In darkness, it may exhibit a resistance of tens or even hundreds of kilo ohms. Under high brightness, the resistance may be as little as a few hundred ohms. One way of using these devices is by placing them in a voltage divider. The resulting voltage will reflect the light levels.

1. First, find the light dependent resistor (LDR, Figure 4). The resistivity of the red material changes with the intensity of light striking it.

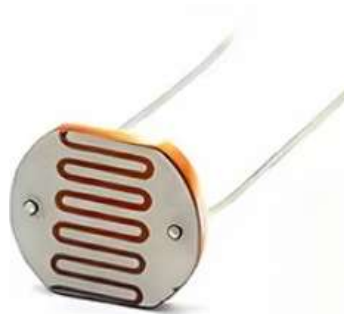


Figure 4 CdS LDR

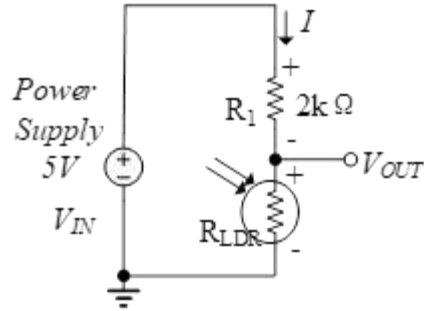


Figure 5

2. Use the DMM to measure the resistance of the LDR.
 - 1) Do this away from any windows and do not block ambient room lighting from hitting the device. Record the resistance value in Table 1 under “ R_{Normal} ”.
 - 2) Place a finger over the LDR to block all light, making sure that the leads are not also touched. Record the resulting resistance value under “ R_{Dark} ”.
 - 3) Shine the pen light directly onto the LDR at a distance of about 25 centimeters. Record the result under “ R_{Bright} ”.
3. Construct the circuit of Figure 5 using $V_{\text{in}}=5\text{V}$ and $R=2\text{k}\Omega$. Using the voltage divider rule, determine the expected value for the voltage across R under normal lighting Dark and Bright conditions.
4. Then your DMM to measure the voltage V_{OUT} under normal lighting Dark and Bright conditions. (12pts,1pt each)

	R_{Normal}	R_{Dark}	R_{Bright}
Resistance of LDR			
$V_{\text{out@Theory}}$			
$V_{\text{out@Experiment}}$			
%Deviation			